

Acids and Bases

MCQ

1. Acids and bases are important because they are used in many _____ processes.

- A) Physical
- B) Chemical
- C) Mechanical
- D) Electrical

Answer: B

2. Citric acid is naturally found in

- A) Milk
- B) Apples
- C) Oranges
- D) Tamarind

Answer: C

3. Tartaric acid is present in

- A) Lemon
- B) Tamarind
- C) Apple
- D) Milk

Answer: B

4. Malic acid is found in

- A) Apples
- B) Oranges
- C) Milk

D) Vinegar

Answer: A

5. Lactic acid is found in

A) Fruits

B) Milk

C) Vegetables

D) Vinegar

Answer: B

6. Hydrochloric acid is present in

A) Blood

B) Saliva

C) Gastric juice

D) Urine

Answer: C

7. Vinegar contains

A) Citric acid

B) Lactic acid

C) Acetic acid

D) Formic acid

Answer: C

8. An acid turns _____ litmus red.

A) Red

B) Blue

C) Green

D) Yellow

Answer: B

9. A base turns _____ litmus blue.

- A) Red
- B) Blue
- C) Yellow
- D) Green

Answer: A

10. A salt is a substance that does not affect

- A) Phenolphthalein
- B) Turmeric
- C) Litmus
- D) Methyl orange

Answer: C

11. Acids donate

- A) Hydroxide ions
- B) Sodium ions
- C) Hydrogen ions
- D) Chloride ions

Answer: C

12. Bases release

- A) Hydrogen ions
- B) Hydroxide ions
- C) Carbon dioxide
- D) Oxygen

Answer: B

13. The word acid comes from the Latin word

- A) Aqua
- B) Acetus

C) Acidus

D) Acris

Answer: C

14. Organic acids mainly come from

A) Minerals

B) Industries

C) Living sources

D) Metals

Answer: C

15. An example of an organic acid is

A) Hydrochloric acid

B) Nitric acid

C) Sulphuric acid

D) Citric acid

Answer: D

16. Inorganic acids are also called

A) Weak acids

B) Natural acids

C) Mineral acids

D) Fatty acids

Answer: C

17. Which acid does not contain carbon

A) Citric acid

B) Tartaric acid

C) Acetic acid

D) Hydrochloric acid

Answer: D

18. Organic acids are generally

- A) Stronger than inorganic acids
- B) Weaker than inorganic acids
- C) Same strength
- D) Neutral

Answer: B

19. Acids are usually _____ in taste.

- A) Sweet
- B) Bitter
- C) Sour
- D) Salty

Answer: C

20. Strong acids are _____ in nature.

- A) Soft
- B) Corrosive
- C) Flexible
- D) Elastic

Answer: B

21. Benzoic acid exists in

- A) Liquid state
- B) Gas state
- C) Solid state
- D) Plasma state

Answer: C

22. Blue litmus in acid turns

- A) Blue
- B) Green

C) Yellow

D) Red

Answer: D

23. Acids are generally

A) Black

B) Coloured

C) Colourless

D) Brown

Answer: C

24. Acid solutions conduct electricity because of

A) Heat

B) Light

C) Ionisation

D) Evaporation

Answer: C

25. Zinc reacts with acid to release

A) Oxygen

B) Nitrogen

C) Hydrogen

D) Carbon dioxide

Answer: C

26. Acids react with metal carbonates to produce

A) Hydrogen

B) Carbon dioxide

C) Nitrogen

D) Oxygen

Answer: B

27. Metal oxides react with acids to form

- A) Salt and hydrogen
- B) Salt and water
- C) Base and water
- D) Acid and salt

Answer: B

28. Which acid helps in digestion

- A) Acetic acid
- B) Citric acid
- C) Hydrochloric acid
- D) Lactic acid

Answer: C

29. Vinegar is used to

- A) Clean metals
- B) Preserve food
- C) Make soap
- D) Produce gas

Answer: B

30. Sulphuric acid is called

- A) Queen of chemicals
- B) King of chemicals
- C) Father of acids
- D) Mother of bases

Answer: B

31. DNA stands for

- A) Deoxyribonucleic acid
- B) Dinitric acid

- C) Dual nucleic acid
- D) Dynamic nucleic acid

Answer: A

32. Bases are generally _____ in taste.

- A) Sour
- B) Sweet
- C) Bitter
- D) Salty

Answer: C

33. Bases are corrosive and may cause

- A) Fever
- B) Blisters
- C) Cold
- D) Allergy

Answer: B

34. Water soluble bases are called

- A) Acids
- B) Salts
- C) Alkalis
- D) Oxides

Answer: C

35. Sodium hydroxide is a

- A) Weak acid
- B) Strong base
- C) Neutral salt
- D) Gas

Answer: B

36. Washing soda is

- A) NaOH
- B) NaCl
- C) Na_2CO_3
- D) NaHCO_3

Answer: C

37. Baking soda is

- A) Sodium chloride
- B) Sodium carbonate
- C) Sodium bicarbonate
- D) Calcium carbonate

Answer: C

38. Bases give soapy touch in

- A) Solid form
- B) Dry state
- C) Aqueous solution
- D) Gaseous state

Answer: C

39. Phenolphthalein turns _____ in base.

- A) Colourless
- B) Red
- C) Pink
- D) Yellow

Answer: C

40. Methyl orange turns yellow in

- A) Acid
- B) Base

C) Salt

D) Water

Answer: B

41. Aluminium reacts with sodium hydroxide to release

A) Oxygen

B) Hydrogen

C) Nitrogen

D) Carbon dioxide

Answer: B

42. Sodium hydroxide reacts with carbon dioxide to form

A) Sodium chloride

B) Sodium carbonate

C) Sodium nitrate

D) Sodium sulphate

Answer: B

43. Bases react with ammonium salts to release

A) Oxygen

B) Hydrogen

C) Ammonia

D) Nitrogen

Answer: C

44. Potassium hydroxide is used to make

A) Washing soap

B) Bathing soap

C) Detergent

D) Fertilizer

Answer: B

45. Calcium hydroxide is used for

- A) Cooking
- B) Painting
- C) White washing
- D) Digestion

Answer: C

46. Antacids contain

- A) Strong acids
- B) Weak acids
- C) Weak bases
- D) Salts

Answer: C

47. Neutralisation reaction forms

- A) Acid and base
- B) Salt and water
- C) Gas and water
- D) Acid and salt

Answer: B

48. HCl reacts with NaOH to form

- A) Sodium sulphate
- B) Sodium nitrate
- C) Sodium chloride
- D) Sodium carbonate

Answer: C

49. Bee sting venom is

- A) Basic
- B) Neutral

C) Acidic

D) Salty

Answer: C

50. Formic acid is injected by

A) Wasp

B) Bee

C) Snake

D) Spider

Answer: B

51. Baking soda neutralizes

A) Base

B) Salt

C) Acid

D) Water

Answer: C

52. Wasp sting venom is

A) Acidic

B) Neutral

C) Alkaline

D) Salty

Answer: C

53. Vinegar is used for wasp sting because it is

A) Base

B) Acid

C) Salt

D) Neutral

Answer: B

54. Tooth decay is caused due to

- A) Bases
- B) Salts
- C) Acids
- D) Water

Answer: C

55. Toothpaste contains

- A) Strong acids
- B) Weak bases
- C) Strong bases
- D) Salts

Answer: B

56. Acidity in stomach is due to excess

- A) Base
- B) Salt
- C) Acid
- D) Water

Answer: C

57. Aluminium hydroxide is an

- A) Acid
- B) Base
- C) Salt
- D) Indicator

Answer: B

58. Acidic soil is treated with

- A) Urea
- B) Lime

C) Water

D) Sand

Answer: B

59. Lime is used in industries to neutralize

A) Basic waste

B) Neutral waste

C) Acidic waste

D) Solid waste

Answer: C

60. Indicators change colour due to change in

A) Temperature

B) Pressure

C) pH

D) Density

Answer: C

61. Litmus is obtained from

A) Flowers

B) Fruits

C) Lichens

D) Roots

Answer: C

62. Turmeric indicator remains yellow in

A) Base

B) Acid

C) Salt

D) Alkali

Answer: B

63. Turmeric turns red in

- A) Acid
- B) Neutral
- C) Base
- D) Water

Answer: C

64. Hibiscus indicator turns deep red in

- A) Base
- B) Acid
- C) Neutral
- D) Salt

Answer: B

65. Hibiscus indicator turns green in

- A) Acid
- B) Base
- C) Salt
- D) Water

Answer: B

66. Beetroot juice is used as a

- A) Base
- B) Acid
- C) Indicator
- D) Salt

Answer: C

67. Red litmus turns blue in

- A) Acid
- B) Base

C) Salt

D) Water

Answer: B

68. Blue litmus turns red in

A) Base

B) Neutral

C) Acid

D) Alkali

Answer: C

69. Phenolphthalein is colourless in

A) Base

B) Salt

C) Acid

D) Alkali

Answer: C

70. Methyl orange turns red in

A) Base

B) Neutral

C) Acid

D) Water

Answer: C

71. Bases conduct electricity in

A) Solid state

B) Gaseous state

C) Aqueous solution

D) Dry state

Answer: C

72. Acids are soluble in

- A) Oil
- B) Alcohol
- C) Water
- D) Kerosene

Answer: C

73. Sodium hydroxide is commonly called

- A) Slaked lime
- B) Caustic soda
- C) Baking soda
- D) Washing soda

Answer: B

74. Calcium hydroxide is called

- A) Quick lime
- B) Slaked lime
- C) Lime stone
- D) Chalk

Answer: B

75. Sodium bicarbonate reacts with acid to release

- A) Oxygen
- B) Hydrogen
- C) Carbon dioxide
- D) Nitrogen

Answer: C

76. Neutral salt does not affect

- A) Turmeric
- B) Phenolphthalein

- C) Litmus
- D) Methyl orange

Answer: C

77. Acids generally exist in

- A) Solid state only
- B) Liquid state
- C) Gaseous state
- D) Plasma state

Answer: B

78. Bases are mostly

- A) Liquids
- B) Gases
- C) Solids
- D) Plasmas

Answer: C

79. Alkalis are bases that are

- A) Insoluble in water
- B) Soluble in water
- C) Weak in nature
- D) Neutral

Answer: B

80. The reaction between acid and base is called

- A) Oxidation
- B) Reduction
- C) Neutralisation
- D) Decomposition

Answer: C

